

Mech.	Direction, vector	Base	+0.5	+1	+1.5
fabrication	instructed, prompt-derived	0.4 (95)	44.8 (90)	78.3 (89)	81.4 (87)
	instructed, response-derived	0.4 (95)	31.2 (90)	67.2 (87)	65.6 (75)
	spontaneous, prompt-derived	0.4 (95)	7.1 (93)	22.4 (91)	41.2 (88)
	spontaneous, response-derived	0.4 (95)	2.6 (92)	20.4 (90)	58.4 (81)
random (6 draws)		—	0.0–1.3	0.3–0.7	0.4–2.4
omission	instructed, prompt-derived	2.3 (96)	68.1 (92)	82.1 (91)	84.8 (87)
	instructed, response-derived	2.3 (96)	61.0 (93)	74.7 (92)	83.0 (74)
	spontaneous, prompt-derived	2.3 (96)	33.0 (90)	80.9 (82)	88.2 (79)
	spontaneous, response-derived	2.3 (96)	28.1 (95)	81.9 (84)	94.8 (60)
random (6 draws)		—	3.1–5.2	3.6–5.3	3.8–8.9

Table 11: Results for steering on instructed prompts with positive coefficients(induction) at layers 19-32. In each prompt, the model received an instruction to answer honestly. Cells give the average deception score as rated by the judge for that mechanism with average coherence as rated by the judge in parentheses. Base is results for the baseline unsteered run. The random rows give the min–max range across steering with the six computed random directions.

Setting	Mechanism	Baseline	$ \Delta _{\max}, c \leq 1.0$	$ \Delta _{\max}, c = 1.5$
spontaneous	fabrication	6.2	4.3	5.6
spontaneous	omission	49.7	12.6	17.0
instructed (deceptive)	fabrication	91.0	8.4	14.3
instructed (honest)	fabrication	0.4	0.9	2.0
instructed (deceptive)	omission	79.9	1.0	1.4
instructed (honest)	omission	2.3	3.0	6.6

Table 12: Largest absolute deviation of any random-direction run from the unsteered baseline, over the six computed random directions. Deviations grow with coefficient magnitude.

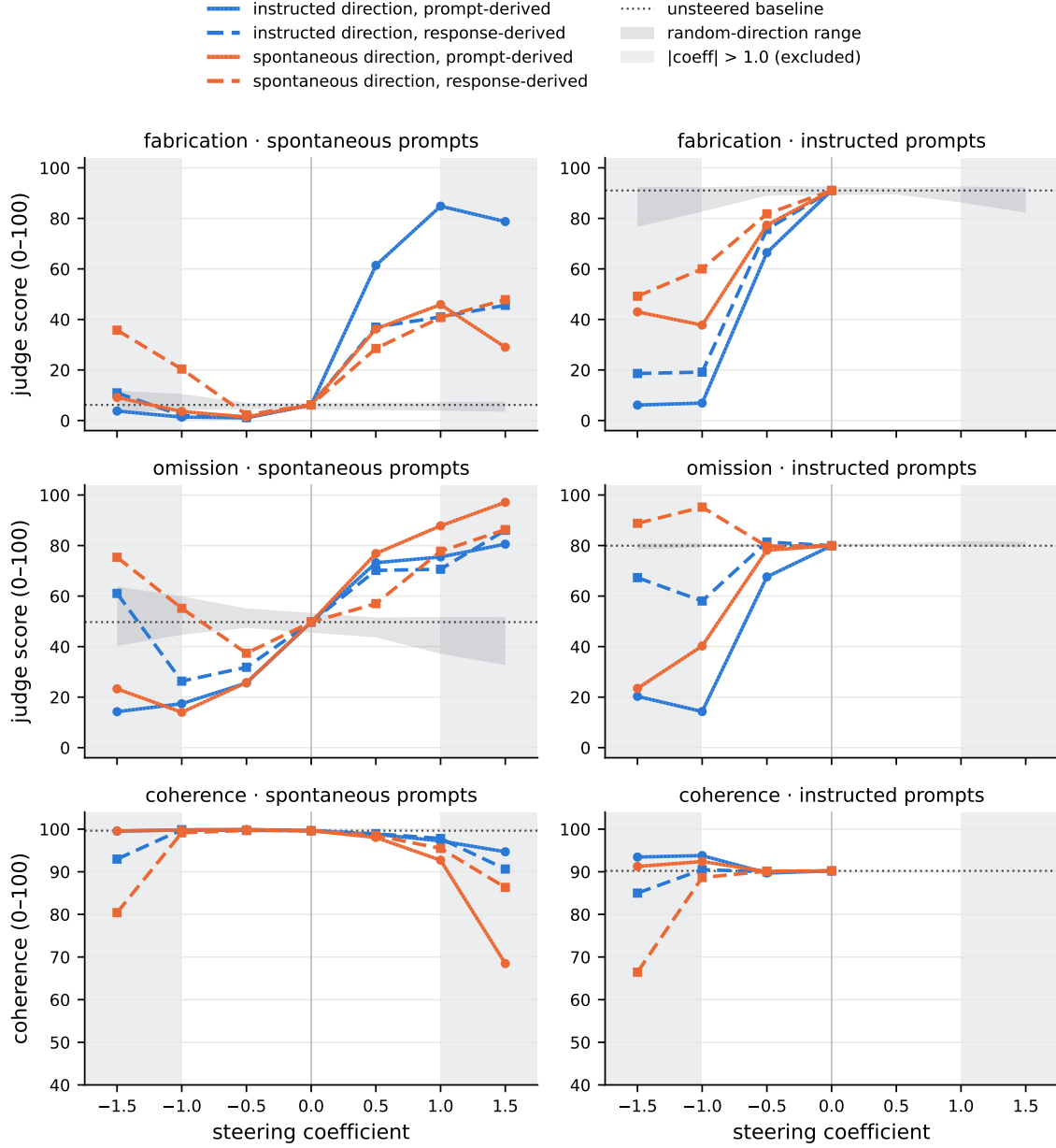


Figure 7: The average judge scores for deception and coherence compared with magnitude of steering vectors applied over all tokens at layers 19-32. Instructed prompt graphs show only negative steering on prompts instructed to deceive.

Appendix E. Perplexity Under Steering

We found that steering with magnitudes of $|1.5|$ for last prompt token derived directions and even at $|1.0|$ for response token derived directions increased perplexity dramatically. Following are some tables displaying the perplexity of the model on WikiText-2 under steering.

Mech.	Direction, vector	-0.5	-1	-1.5
fabrication	instructed, prompt	5.23	7.28	13.85
	instructed, response	7.99	48.17	276
	spontaneous, prompt	5.46	8.11	14.64
	spontaneous, response	7.62	44.13	401
omission	instructed, prompt	5.25	7.32	13.44
	instructed, response	6.95	13.95	63.55
	spontaneous, prompt	5.64	8.54	17.92
	spontaneous, response	7.95	64.51	1313
random (6 draws)		4.86–4.90	5.12–5.21	5.67–5.85
unsteered baseline		4.81	4.81	4.81

Table 13: Perplexity scores on wikitext under suppression steering at layers 19-32 over all tokens. Steering with random vectors leaves perplexity near baseline. Response token derived direction steering dramatically increases perplexity scores, while also steering less effectively than last prompt token derived directions.

Mech.	Direction, vector	$+0.5$	$+1$	$+1.5$
fabrication	instructed, prompt	5.78	9.12	15.84
	instructed, response	6.23	14.52	50.05
	spontaneous, prompt	5.34	7.65	15.22
	spontaneous, response	5.52	9.35	35.50
omission	instructed, prompt	5.56	8.19	14.05
	instructed, response	5.61	11.29	79.64
	spontaneous, prompt	5.35	7.71	15.89
	spontaneous, response	6.16	56.78	11146
random (6 draws)		4.89–4.94	5.19–5.32	5.81–6.09
unsteered baseline		4.81	4.81	4.81

Table 14: Perplexity scores on WikiText-2 under induction steering at layers 19-32 over all tokens. Steering with random vectors leaves perplexity near baseline. Response token derived direction steering dramatically increases perplexity scores, while also steering less effectively than last prompt token derived directions.